[redacted]

→ Recap - Interest Rate Parity (IRP)
- Purchasing Power Parity (PPP)
- International Fisher Effect (IFE)

→ Practice questions on PPP and IFE
→ Discuss Big Mac Index (PPP)
→ Sum up and looking forward.

Interest Rate Parity → interest rate differential between two countries explains premium on exchange rate forward contracts

$$i_h - i_f = p = \frac{F}{S} - 1$$

if $p > i_h - i_f \Rightarrow$ CIA

I - inflation
i - interest (nominal)

Purchasing Power Parity ⇒

$$\%\Delta FX = I_{infl.h} - I_{infl.f}$$
$$= \%\Delta S$$

International Fisher Effect ⇒ $\%\Delta FX = i_h - i_f$

nominal interest rate = inflation + real interest (assumed to be the same among countries)

if expected real interest rate = 2%

$$i_h - i_f = I_h + \text{real rate}_h - (I_f + \text{real rate}_f) =$$
$$= I_h + 2\% - I_f - 2\% = I_h - I_f$$

PPP

→ A pair of skis in the USA costs $1,000

→ If CAN: $0.6667/CAN, how much should that same pair of skis cost in Canada?

$1,000 ⅀ $0.667/CAN

1,000/0.6667 ≈ CAN 1,500.

→ what if prices have not changed in CAN, but ski prices in the US went up to $1,200 USN

ΔFX = 0 (no changes to exchange rates)

→ where would you prefer to purchase these skis?

Cost of skis in Canada in USD = $1,000

Cost of skis in USA = $1,200

You want to purchase skis in Canada as long as price in Canada is less than in the US

$$\%\Delta FX = \frac{1+I_h}{1+I_f}$$

$$PPP: \%\Delta FX = I_h - I_f$$

how that'll adjust in real numbers

All in USD	Cost of skis $t=0$	$t=1$	Inflation
Canada	$1,000	$1,000	$I_{CAN} = \frac{\$1,000}{\$1,000} - 1 = 0$
USA	$1,000	$1,200	$I_{USA} = \frac{\$1,200}{\$1,000} - 1 = 20\%$

According to PPP: $\%\Delta FX = I_{USA} - I_{CAN} = 20\% - 0\% = 20\%$

$\%\Delta FX = \$0.66667/CAN - (+20\%) = \$0.80/CAN$

(3) 11-6-17

Real interest are expected to hover around 2% across the world and nominal interest rates are 13% in USA and 8% in Canada, how would you expect exchange rates to change?

$r = 2\%$ – real interest rate

$i_{USA} = 13\%$, $i_{CAN} = 8\%$

From Canada's perspective

IFE: $\%\Delta FX = i_{CAN} - i_{USA} = 8\% - 13\% = -5\%$

∴ we can also infer that inflation in USA is higher than in Canada because $\%\Delta FX = i_{CAN} - i_{USA} = i_{CAN} + r_{CAN} - i_{USA} - r_{USA}$

What will be the new spot exFX if $S_0 = CAN\ 1.50/\$$

$$S_1 = (1 + \%\Delta FX) \cdot S_0$$

$$= (1 + (-5\%)) \cdot CAN\ 1.50/\$ =$$

$$= 0.95 \cdot 1.50/\$ = CAN\ 1.425/\$$$

Going forward:

Ch. 9. Forecasting next week

Quiz 3 – ch. 8 due Nov 8